Chapter 8

Backbones

Chapter 8: Outline

Backbone Networks

- High speed networks linking an organization's LANs
  - Making information transfer possible between departments
  - Use high speed circuits to connect LANs
  - Provide connections to other backbones, MANs, and WANs
- Sometimes referred to as
  - An enterprise network
  - A campus-wide network

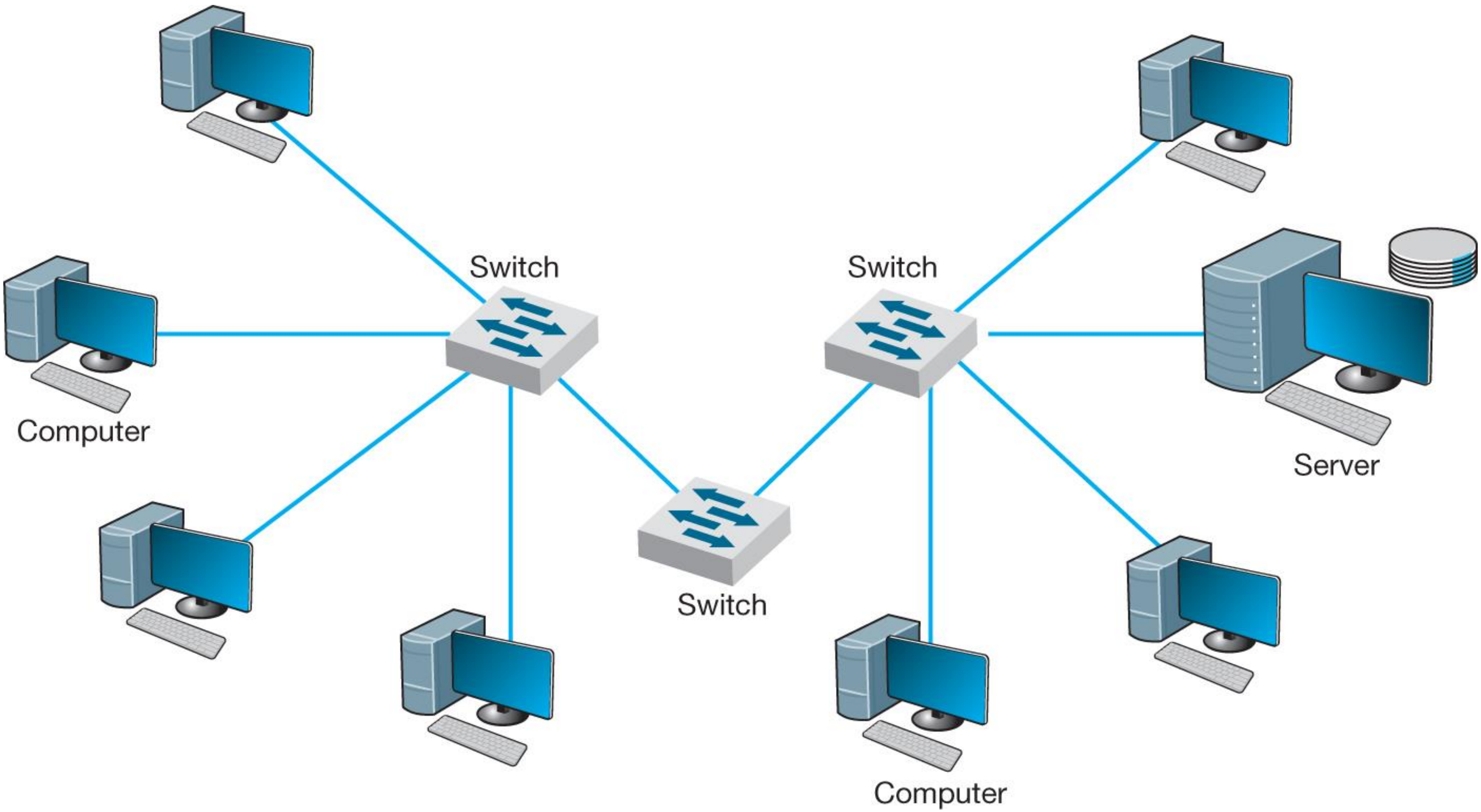
8.2 Backbone Network Components

- Network cable
  - Functions in the same way as in LANs
  - Optical fiber - more commonly chosen because it provides higher data rates
- Hardware devices
  - Computers or special purpose devices used for interconnecting networks
    - Switches
    - Routers
    - Gateways

Backbone Network Devices

Device	Operates At	Packets	Physical Layer	Data Link Layer	Network Layer
Switch	Data link layer	Filtered using data link layer addresses	Same or different	Same	Same
Router	Network layer	Routed using network layer addresses	Same or different	Same or different	Same
Gateway	Network layer	Routed using network layer addresses	Same or different	Same or different	Same or different

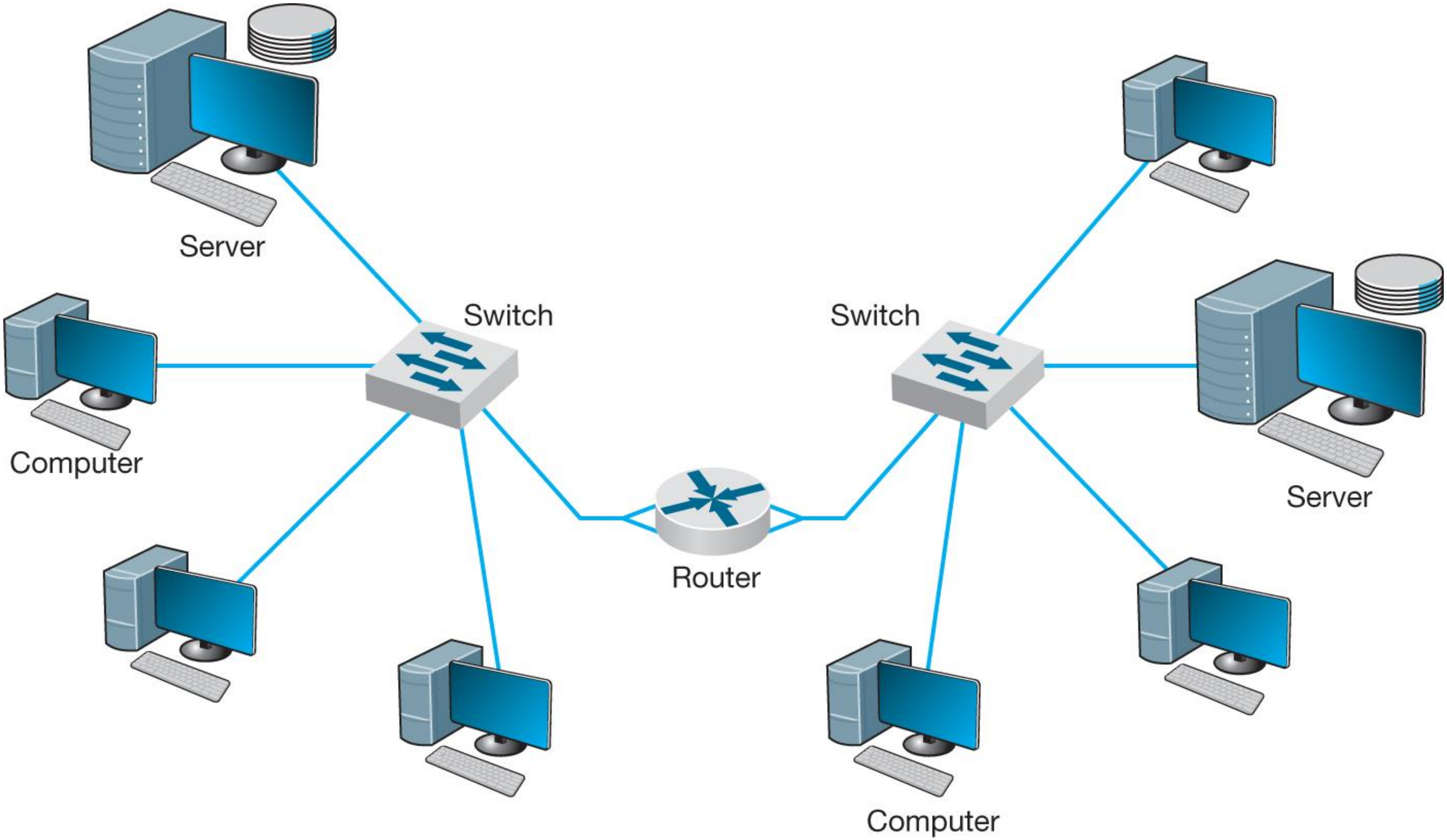
Switches



Switches

- Most switches operate at the data link layer
- They connect two or more network segments that use the same data link and network protocol
- They may connect the same or different types of cable
- These use the data link layer address to forward packets between network segments

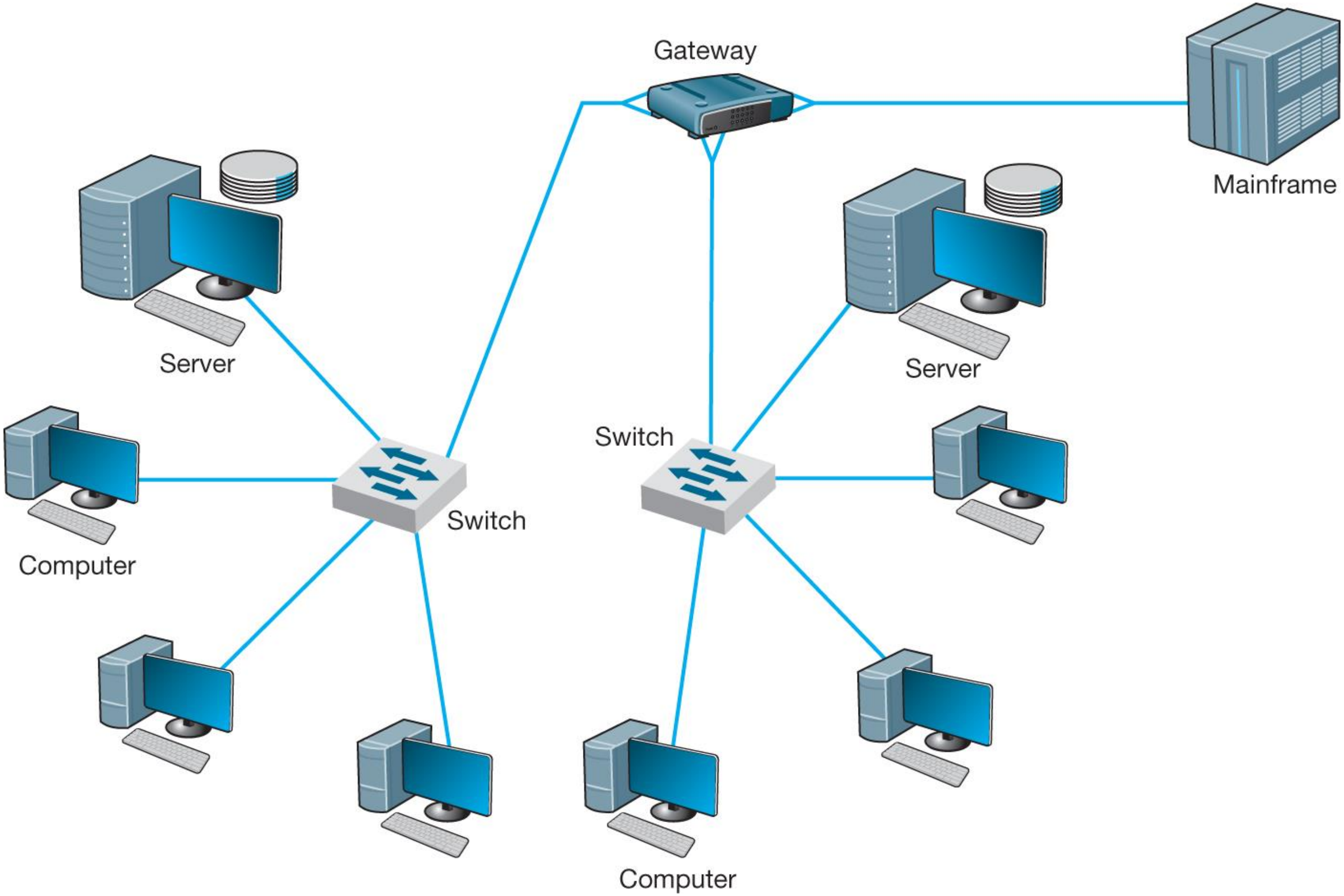
Routers



Routers

- **Operations**
  - Operates at the network layer
  - Examines the destination address of the network layer
  - Strips off the data link layer packet
  - Chooses the “best” route for a packet (via routing tables)
  - Forwards only those messages that need to go to other networks
- **Compared to Switches**
  - Performs more processing
  - Processes only messages specifically addressed to it
  - Recognizes that message is specifically addressed to it before message is passed to network layer for processing
  - Builds new data link layer packet for transmitted packets

Gateways


Gateway
Mainframe
Server
Server
Switch
Switch
Computer
Computer

Gateways

- Operate at network layer and use network layer addresses in processing
- More complex than switches or routers
- Connect two or more networks that use the same or different data link and network protocols
- Some work at the application layer
- Process only those messages addressed to them

Other Backbone Network Devices

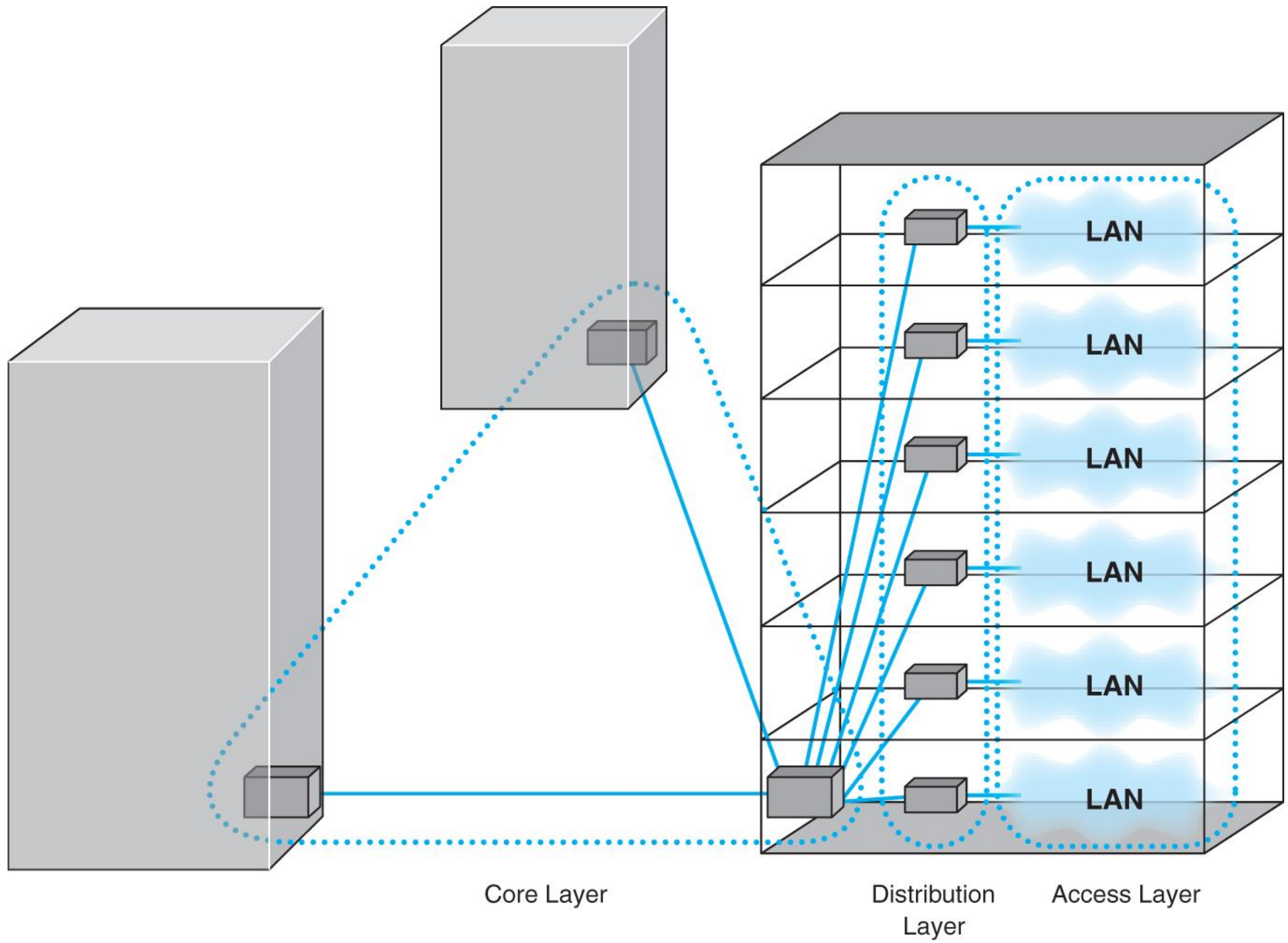
- **Terminology in marketplace is variable by vendor**

- **Layer-3 switches**
  - **Similar to L2 switches, but switch messages based on network layer addresses (usually IP address)**
  - **Have the best of both switches and routers**
  - **Can support more simultaneously active ports than routers**

8.3 Backbone Network Architectures

- **Identifies the way backbone interconnects LANs**
- **Manages way packets from one network move through the backbone to other networks**
- **Three layers:**
  1. ***Access layer:*** **used in LANs attached to BB**
  2. ***Distribution layer:*** **connects LANs together**
  3. ***Core layer:*** **connects different backbone networks together in enterprise network**

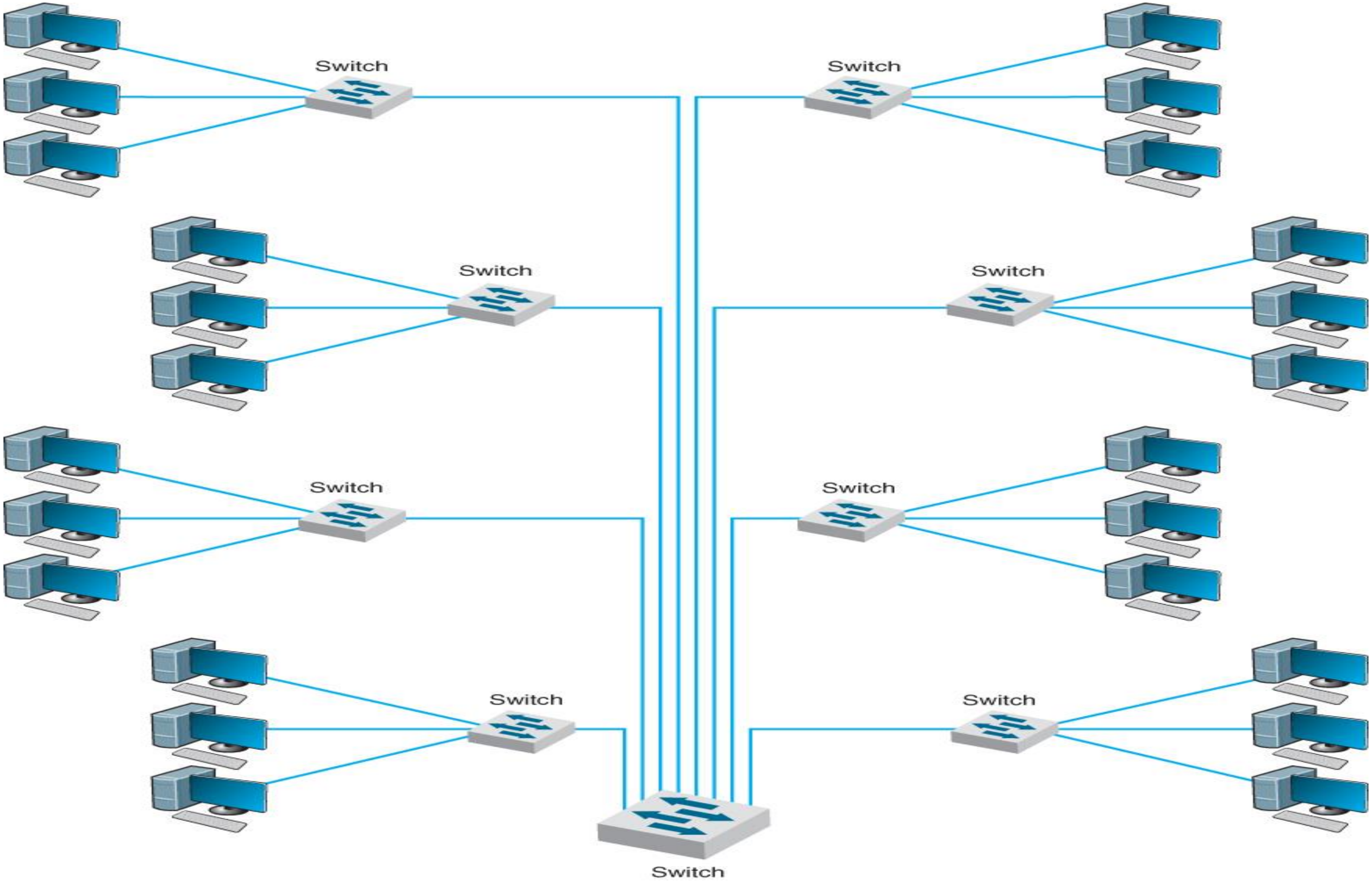
Backbone Network Design Layers



Fundamental Backbone Architectures

- ***Switched Backbones*: most common type of backbone, used in distribution layer, used in new buildings, sometimes in core layer, can be rack or chassis based.**
- ***Routed Backbones*: move packets along backbone on basis of network layer address, typically using bus, Ethernet 100Base-T, sometimes called subnetted backbone**
- ***Virtual LANs*: networks in which computers are assigned into LAN segments by software rather than by hardware; can be single switch or multiswitch VLANs. Very popular technology.**

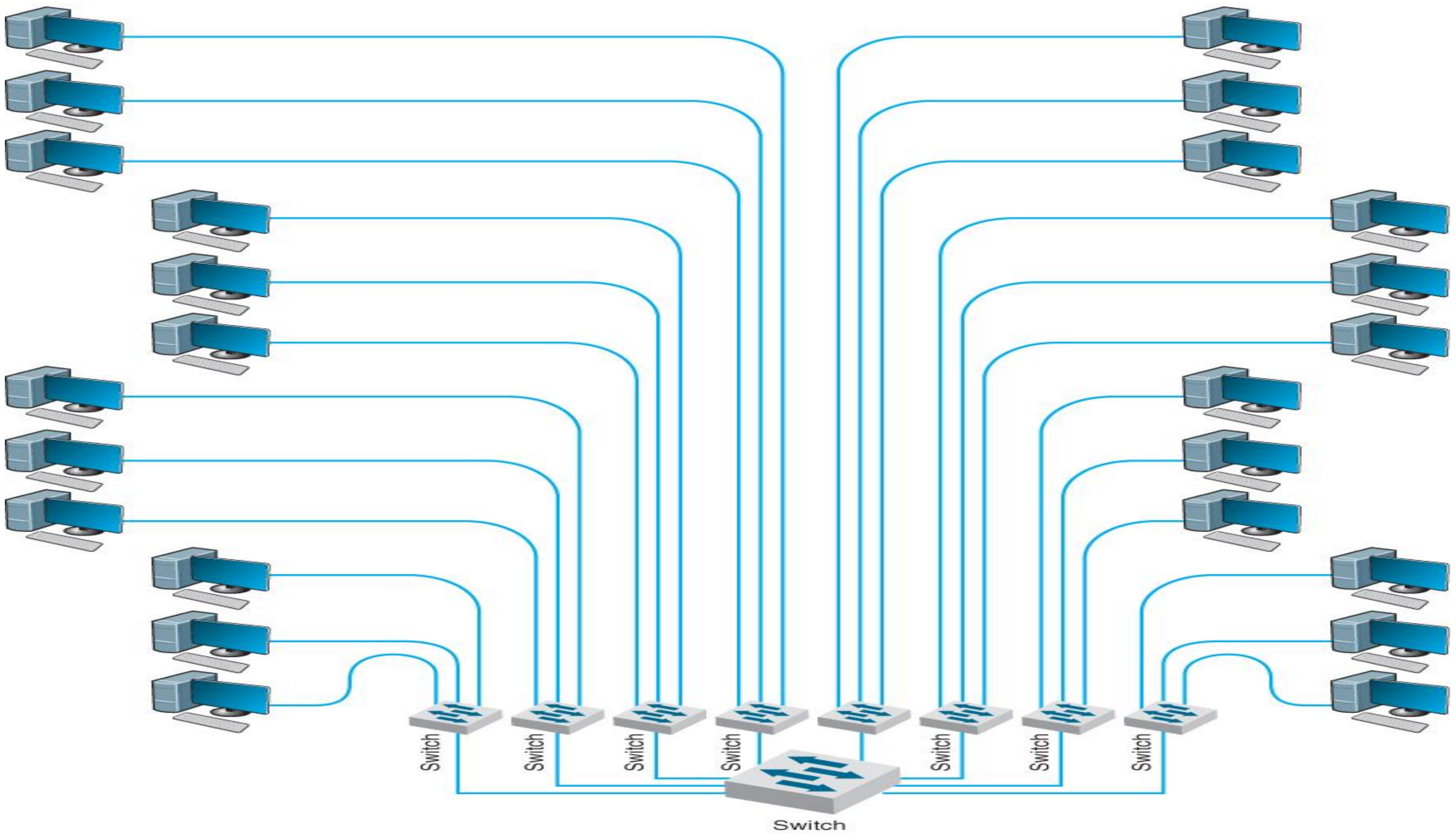
Switched Backbone



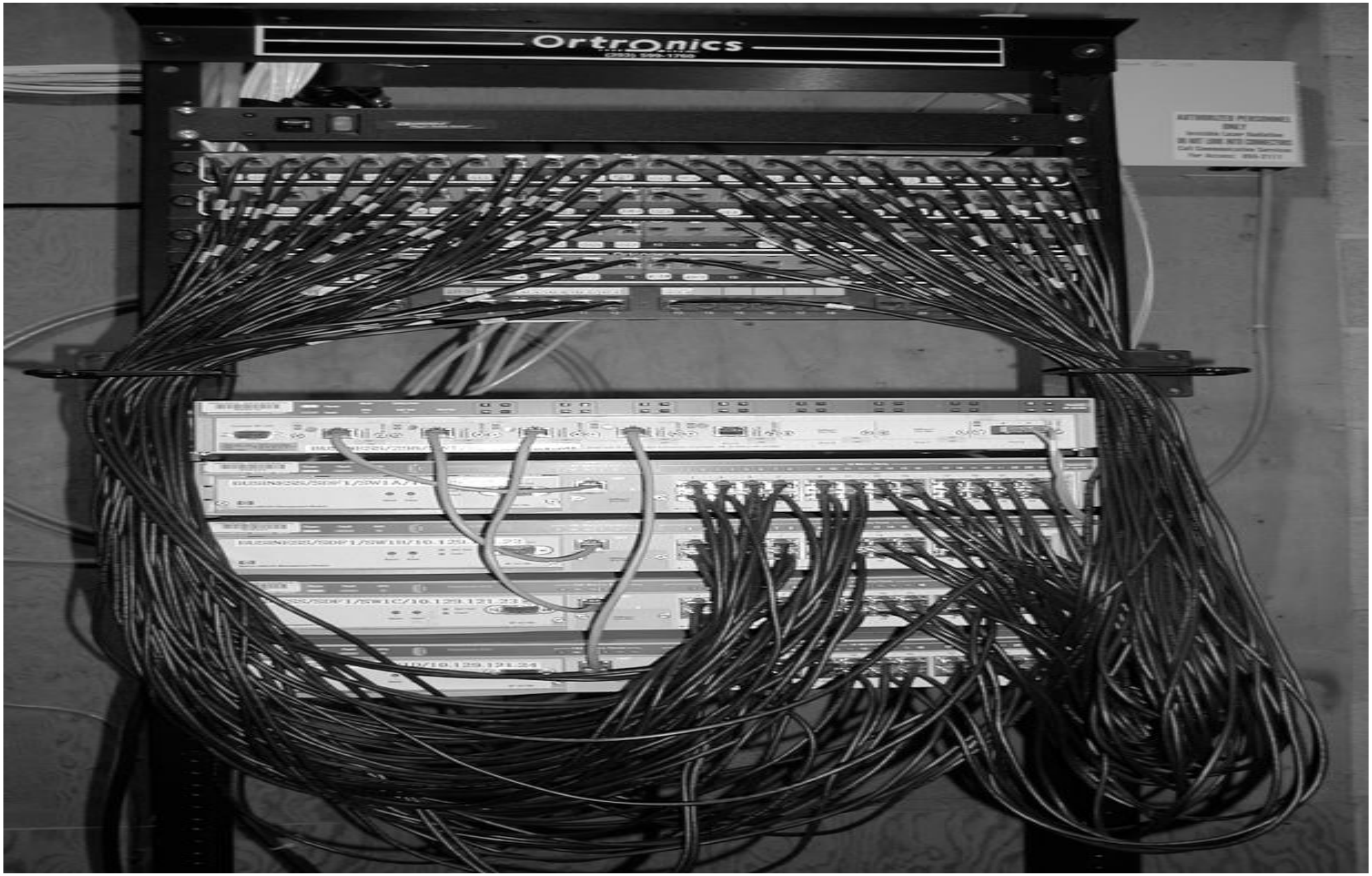
Switched Backbones

- Replaces the many routers of other designs
  - Backbone has more cables, but fewer devices
  - No backbone cable used; switch is the backbone.
- Advantages:
  - Improved performance (200-600% higher) due to simultaneous access of switched operations
  - A simpler more easily managed network – less devices

Rack-Mounted Switched Backbones



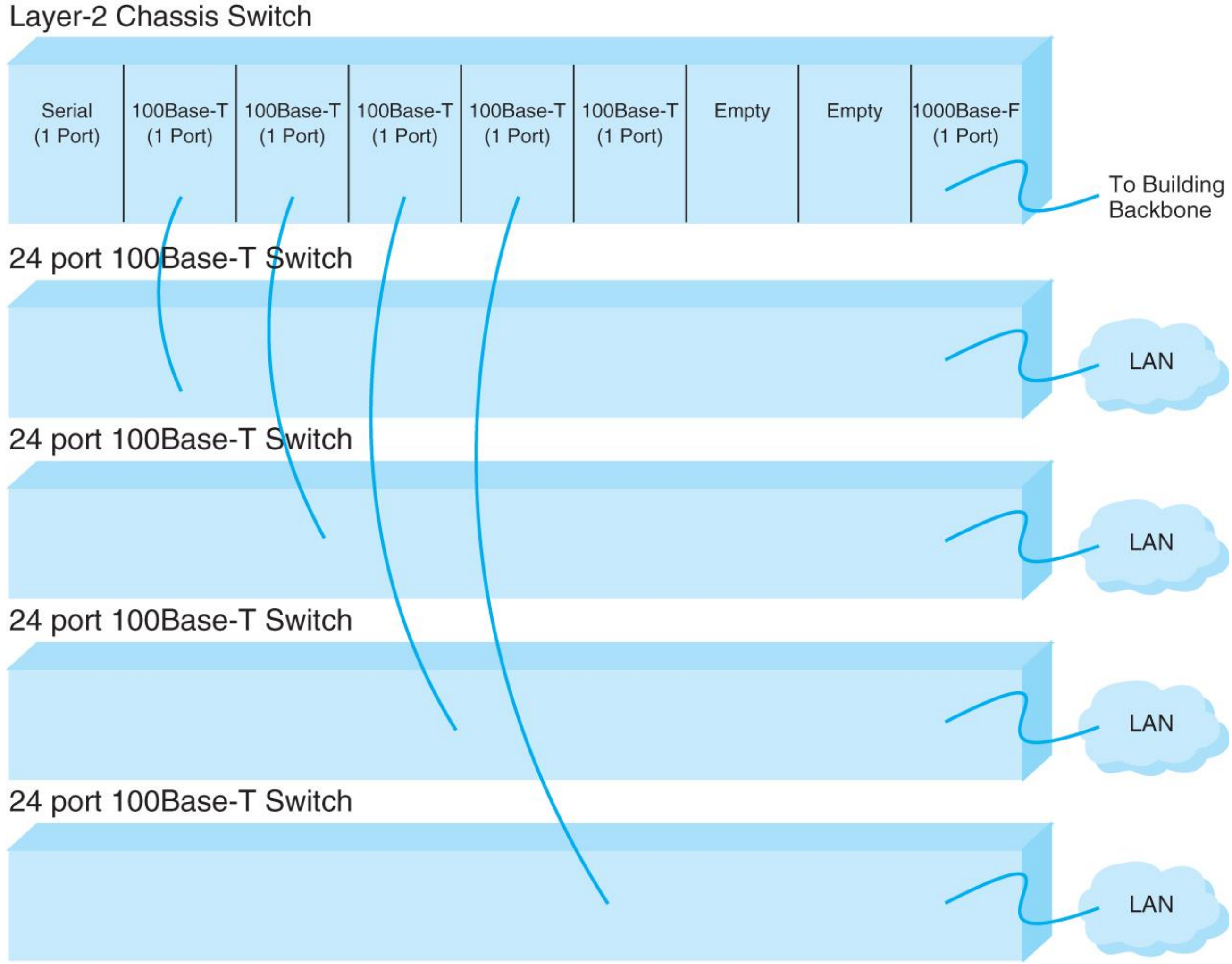
Rack-Mounted Switched Backbones



Rack-Based Switched Backbones

- Places all network switch equipment physically in one "rack" room
  - Easy maintenance and upgrade
  - Requires more cable, but usually small part of overall cost
- Main Distribution Facility (MDF) or Central Distribution Facility (CDF)
  - Another name for the rack room
  - Place where many cables come together
  - Patch cables used to connect devices on the rack
- Easier to move computers among LANs

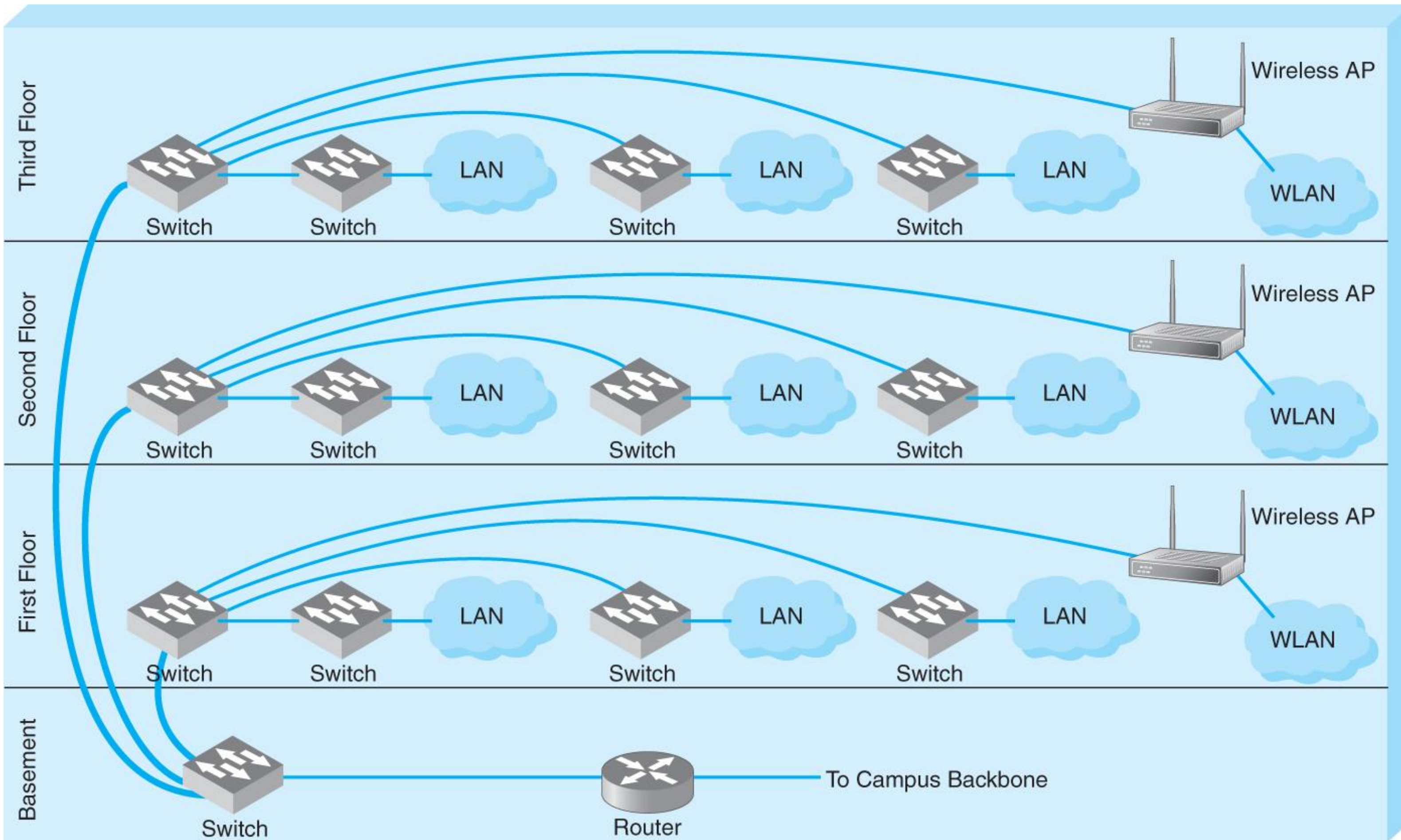
Main Distribution Facility (MDF)



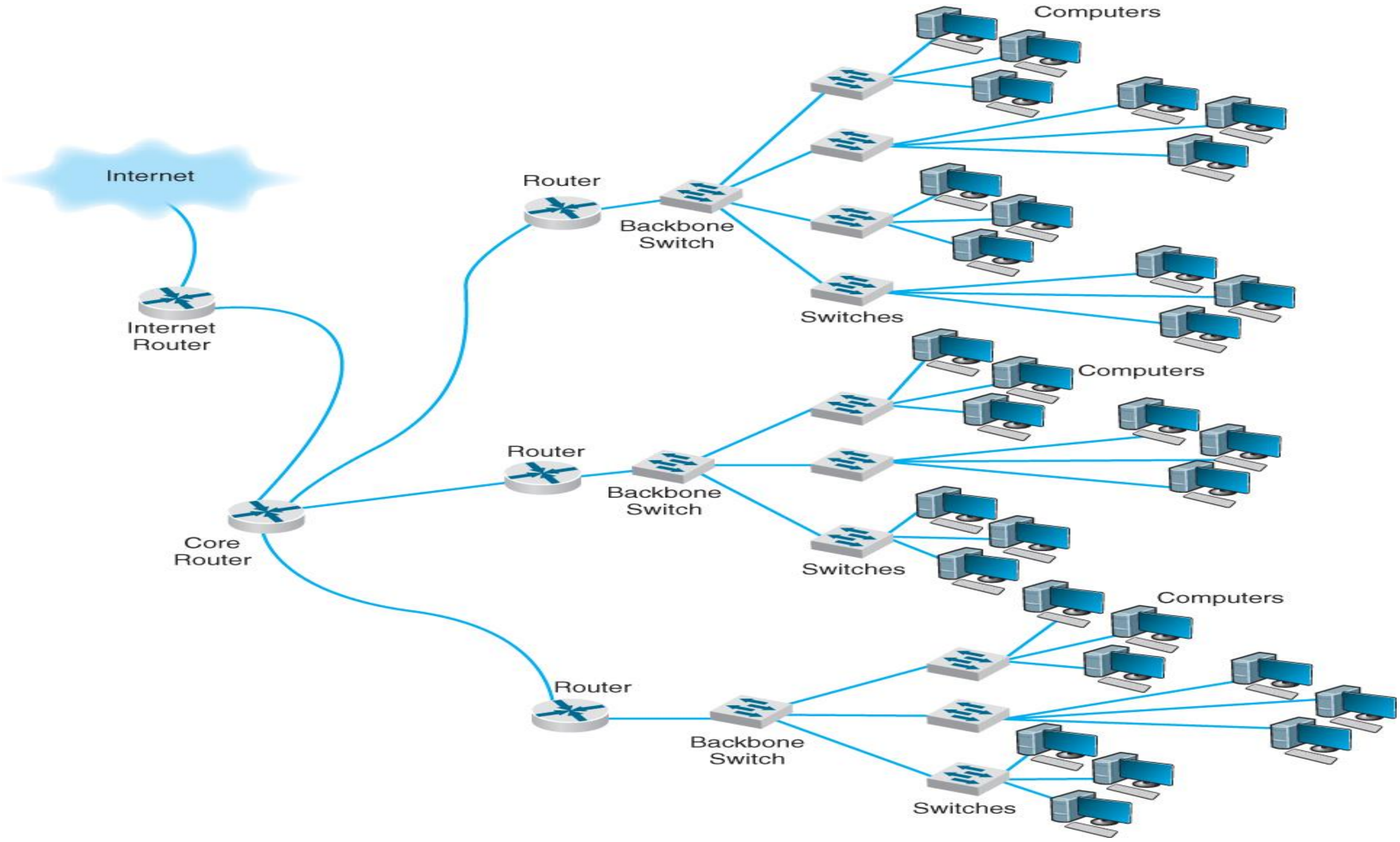
Chassis-Based Switched Backbones

- Use a "chassis" switch instead of a rack
  - Enables administrators to plug modules into switch
  - Modules can vary in nature, router or 4-port 100Base T switch
    - Example of a chassis switch with 710 Mbps capacity
      - 5 10Base-T hubs, 2 10Base-T switches (8 ports each)
      - 1 100Base-T switch (4 ports), 100Base-T router
      - → (5 x 10) + (2 x 10 x 8) + (4 x 100) + 100 = 710 Mbps
- Advantage is flexibility
  - Enables users to plug modules directly into the switch
  - Simple to add new modules

Switched Backbone at Indiana Univ.



Routed Backbone



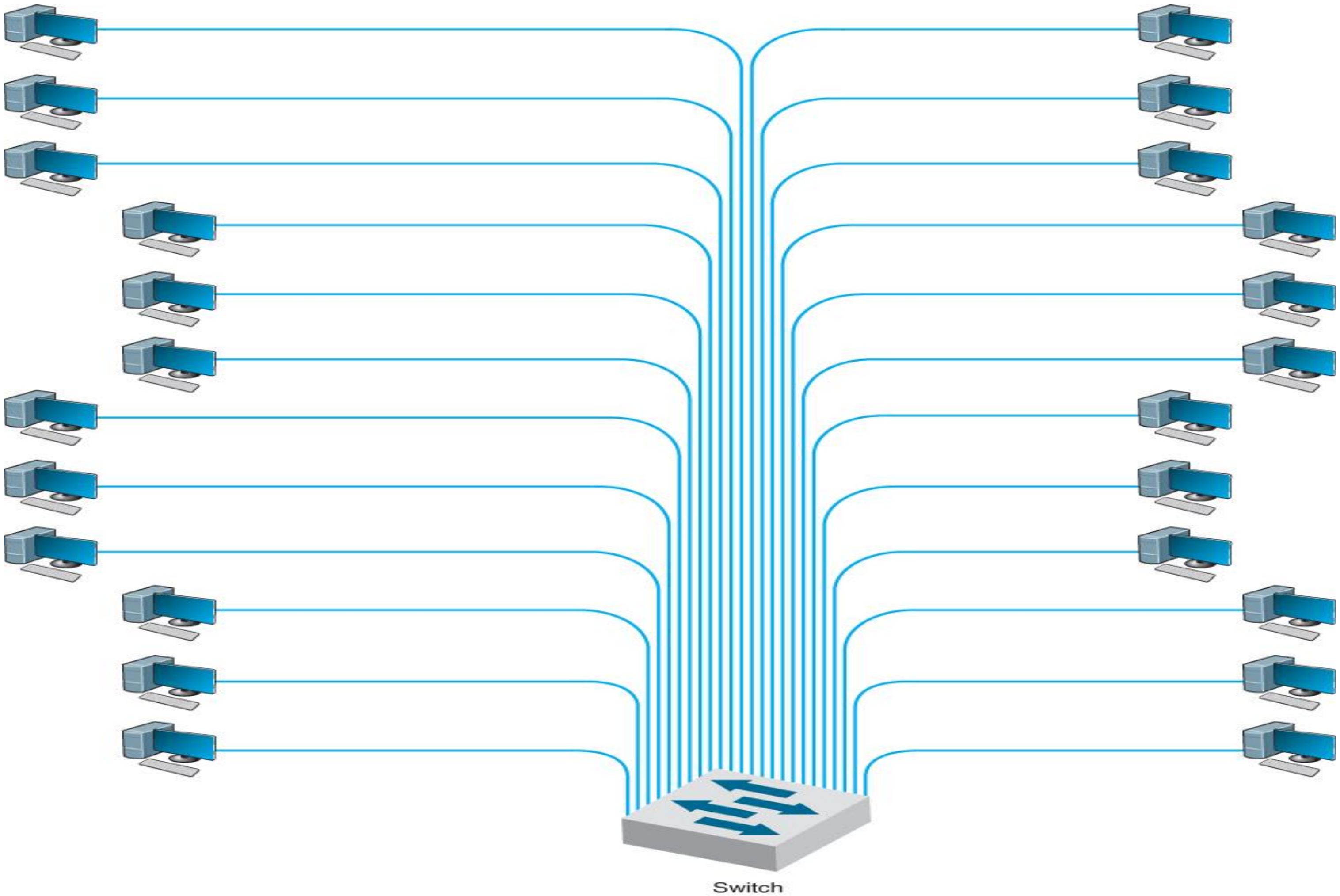
Routed Backbones

- Move packets using network layer addresses
- Commonly used at the core layer
  - Connecting LANs in different buildings in the campus
  - Can be used at the distribution layer as well
- LANs can use different data link layer protocols
- Main advantage: LAN segmentation
  - Each message stays in one LAN; unless addressed outside the LAN
  - Easier to manage, LANs are separate entities, segments
- Main disadvantages
  - Tend to impose time delays
  - Require more management than switches

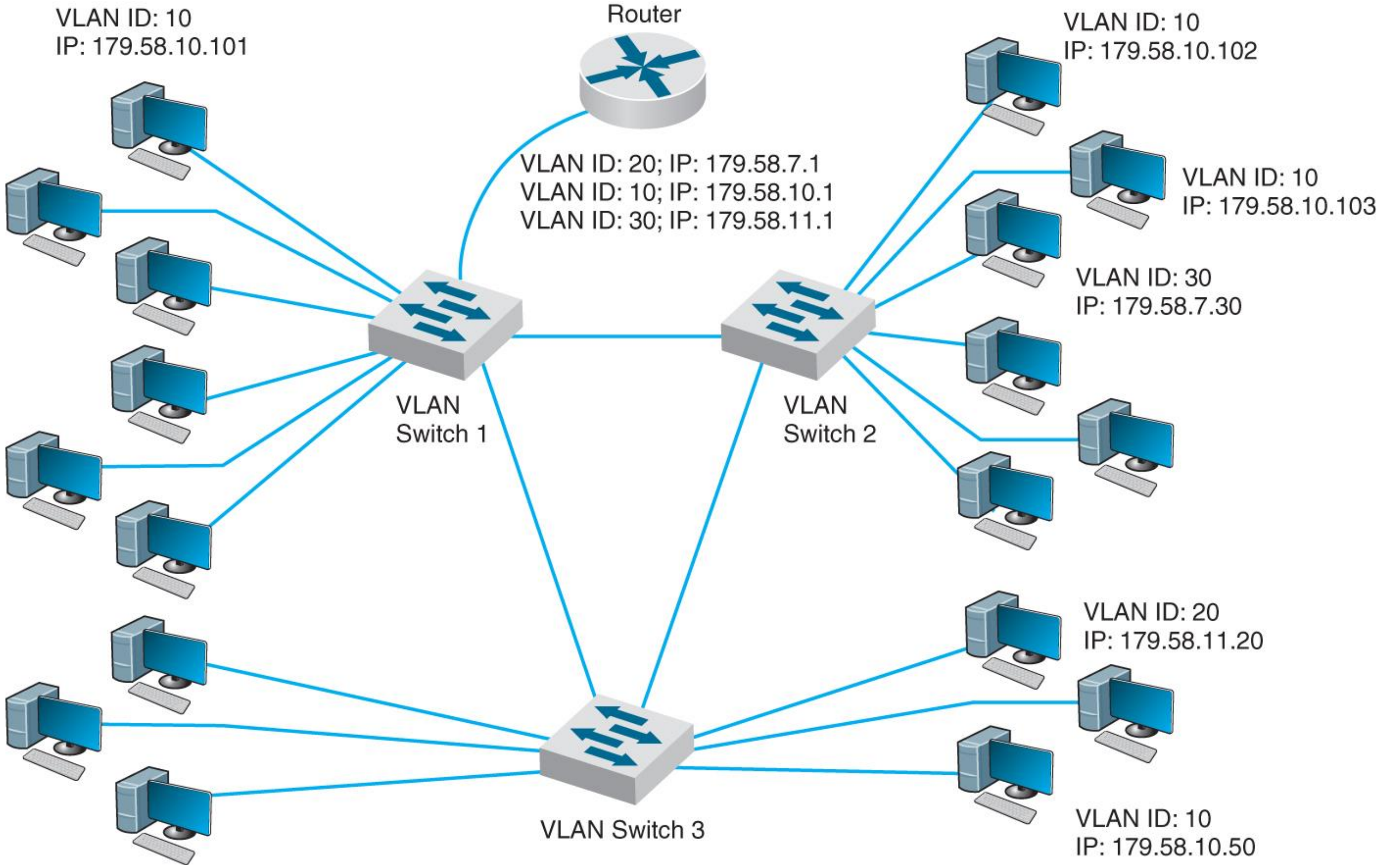
Virtual LANs (VLANs)

- A new type of LAN-BN architecture
  - Made possible by high-speed intelligent switches
  - Computers assigned to LAN segments by software
- Often faster and provide more flexible network management
  - Much easier to assign computers to different segments
- More complex and so far usually used for larger networks
- Basic VLAN designs:
  - Single switch VLANs
  - Multi-switch VLANs

VLAN-based Backbone



Multi-switch VLAN-Based Backbone



How VLANs Work

- **Each computer is assigned into a VLAN that has a VLAN ID**
- **Each VLAN ID is matched to a traditional IP subnet**
  - **Each computer gets an IP address from that switch**
    - **Similar to how DHCP operates**
- **Computers are assigned into the VLAN based on physical port they are plugged into**

Multiswitch VLAN Operations

- Same as single switch VLAN, except uses several switches, perhaps in core between buildings
- Inter-switch protocols
  - Must be able to identify the VLAN to which the packet belongs
- Use IEEE 802.1q (an emerging standard)
  - When a packet needs to go from one switch to another
    - 16-byte VLAN tag inserted into the 802.3 packet by the sending switch
  - When the IEEE 802.1q packet reaches its destination switch
    - Its header (VLAN tag) stripped off and Ethernet packet inside is sent to its destination computer

VLAN Operating Characteristics

- **Advantages of VLANs**
  - **Faster performance: Allow precise management of traffic flow and ability to allocate resources to different type of applications**
  - **Traffic prioritization (via 802.1q VLAN tag)**
    - **Include in the tag: a priority code based on 802.1q**
    - **Can have QoS capability at MAC level**
      - **Similar to RSVP and QoS capabilities at network and transport layers**
- **Drawbacks**
  - **Cost**
  - **Management complexity**
  - **Some “bleeding edge” technology issues to consider**

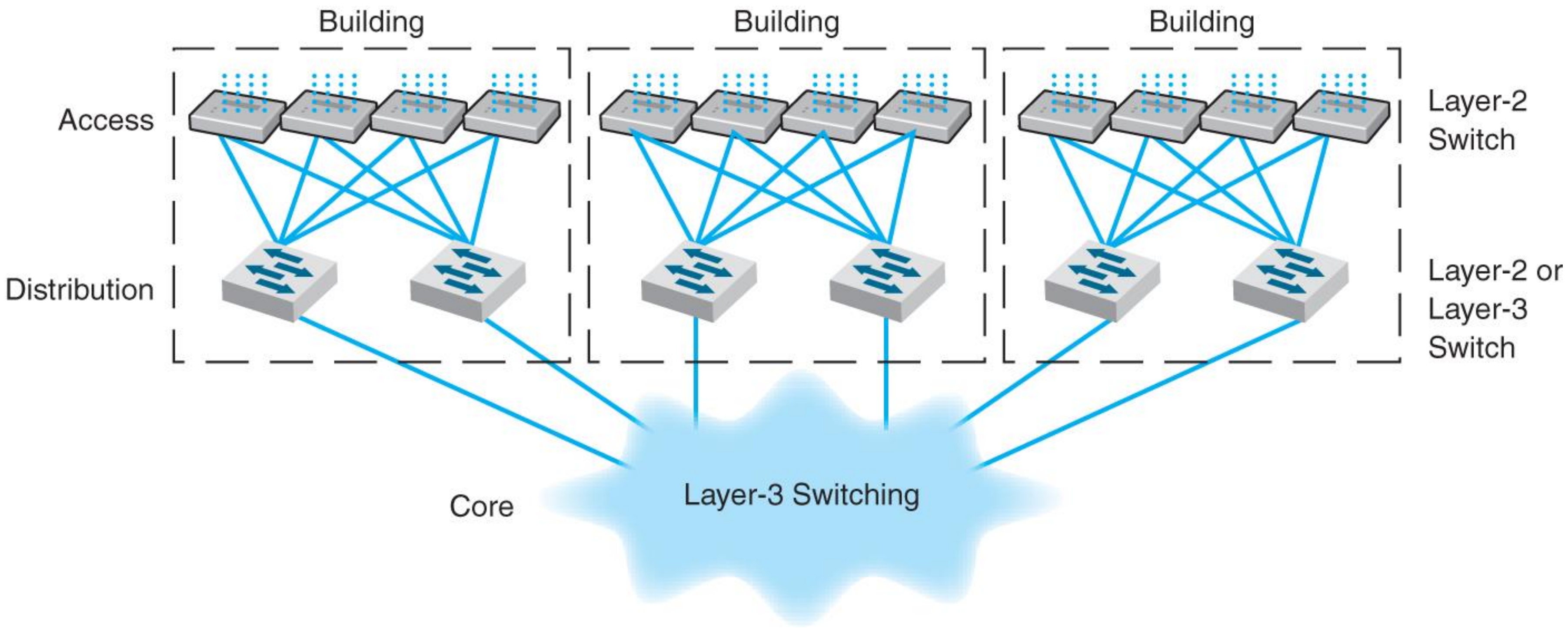
8.4 Best Practice Backbone Design

- Architectures
  - At distribution layer → switched backbone because of performance and cost
  - At core layer → routed backbone
  - VLANs closer but more costly and complex

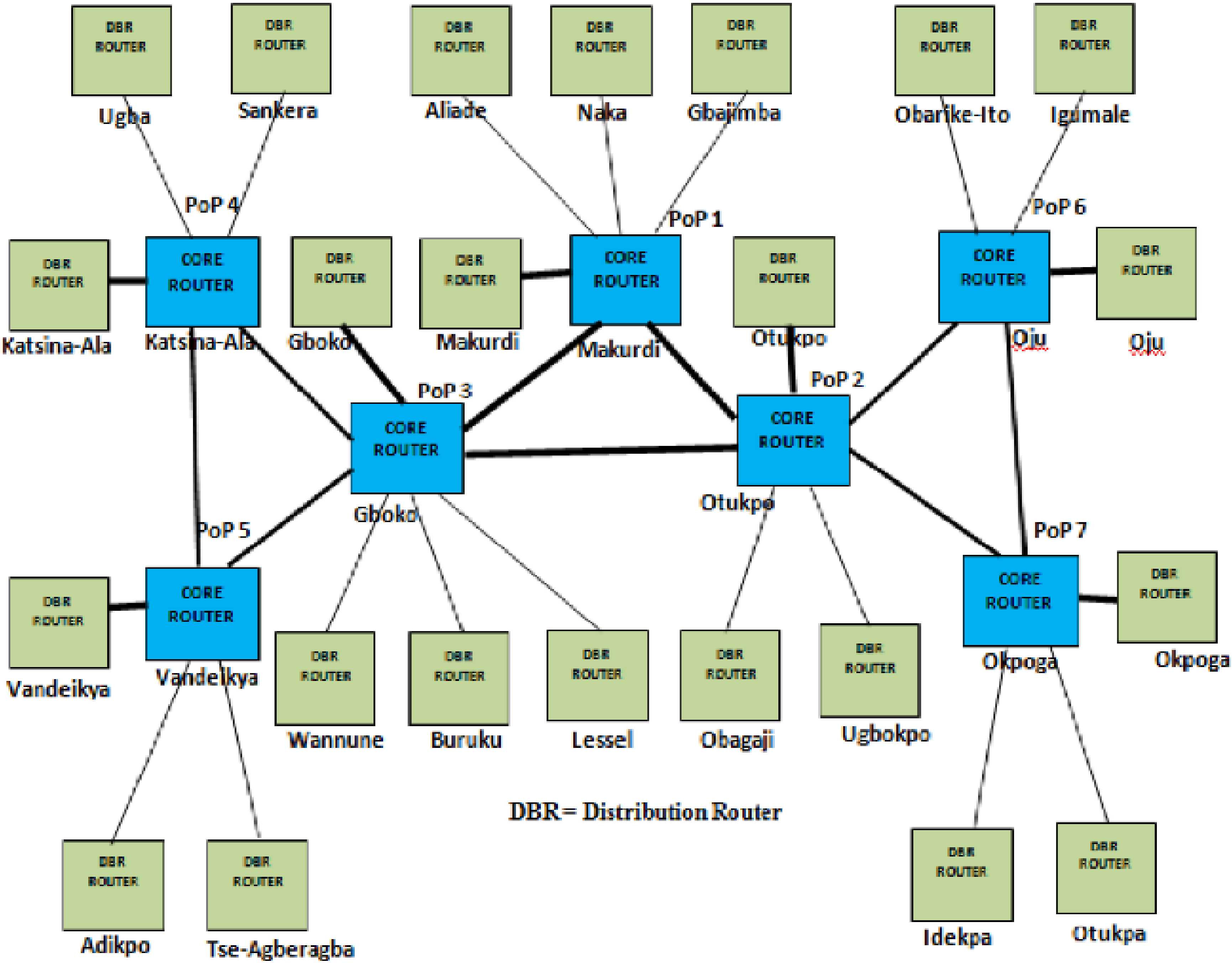
Recommendations for BB Design

- **Best architecture**
  - **Switched backbone or VLAN at distribution layer**
  - **Routed backbone at core layer**
- **Best technology - Gigabit Ethernet**
- **Ideal design**
  - **A mixture of layer-2 and layer-3 Ethernet switches**
  - **Access Layer**
    - **100Base-T Later 2 switches with cat5e or cat6**
  - **Distribution Layer**
    - **100base-T or 1000BaseT/F Layer 3 switches**
  - **Core Layer**
    - **Layer 3 switches running 10GbE or 40GbE over fiber**

Best Practice Network Design [Backup]



Large Enterprise Backbone



DBR = Distribution Router

Layer-3 Switch vs Router

Features	Layer 3 Switch	Router
Layer 3 basic routing	Yes	Yes
Traffic management	Yes	Yes
WAN Interface Card, WIC, support	No	Yes
Advance routing features	No	Yes
Forwarding Architecture	Hardware-ASIC	Software-IOS
Remote Monitoring, RMON, support	Yes	No
Policy performance	High	Low
WAN support	No	Yes
WAN interfaces	No	Yes
QoS features	No	Yes
NAT	No	Yes
LAN capabilities	Yes	No

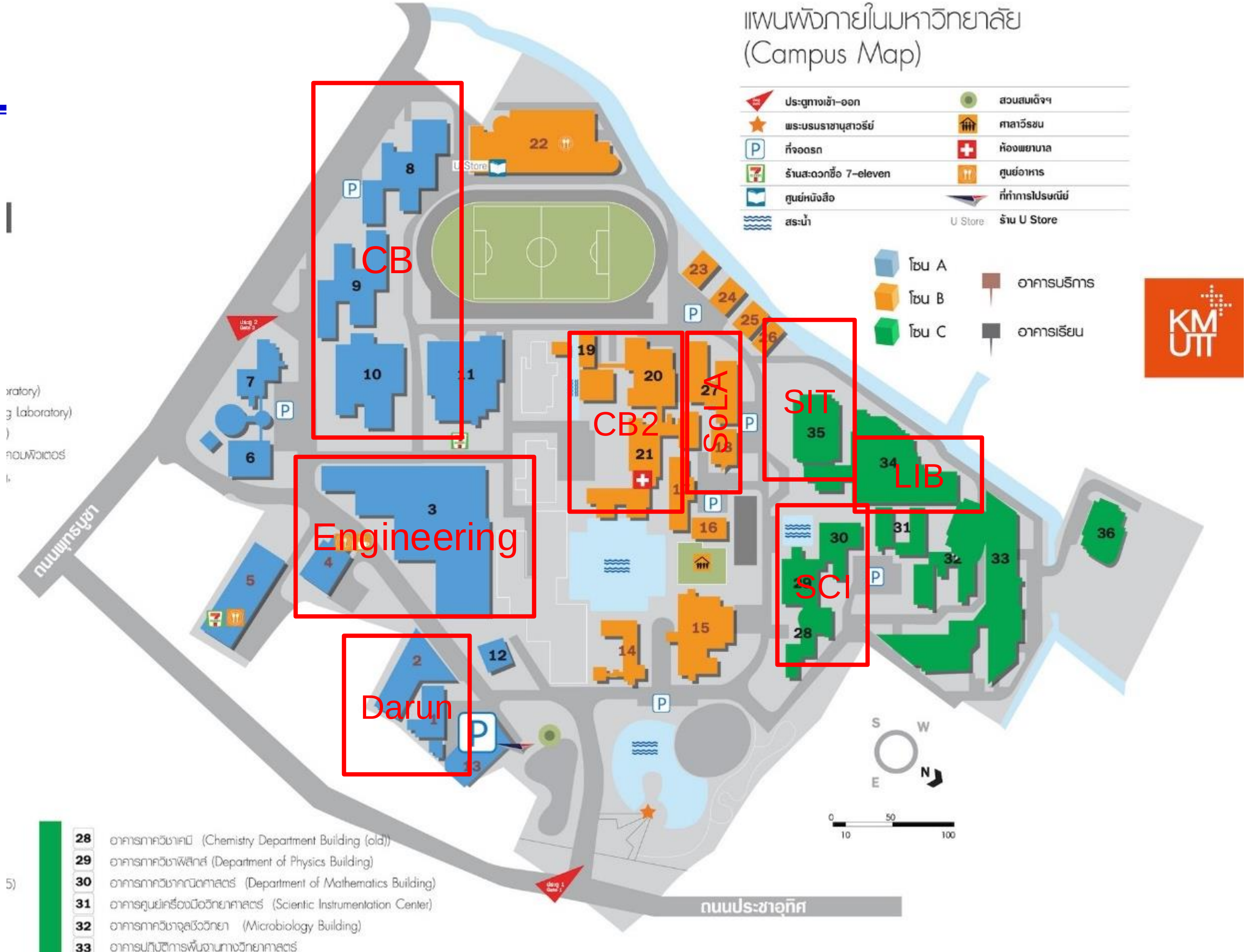
https://community.fs.com/blog/layer-3-switch-vs-router-what-is-your-best-bet.html

8.5 Improving Backbone Performance

- Improve computer and device performance
  - Upgrade them to faster devices
  - Change to a more appropriate routing protocol
    - Distance vector – typically used on BNs
    - Link state – typically used on WANs and MANs
  - Use gigabit Ethernet as BB (eliminate translations)
  - Increase memory in devices
- Improve circuit capacity
  - Upgrade to a faster circuit; Add additional circuits
  - Replace shared circuit BB with a switched BB
- Reduce network demand
  - Restrict applications that use a lot of network capacity
  - Reduce broadcast messages (placing filters at switches)

8.6 Implications for Cyber Security

- Most routers now have software that enables the manager to create an access control list (ACL).
  - Traffic should be allowed or blocked by checking against the ACL rule.
    - Source and destination IP addresses and the action to permit to entry or not by rules of ACL
  - ACL software that enables ACL to have different rules on different interfaces of the router.
  - Some ACL software can specify rules for application layer such as HTTP, SMTP, FTP
- VLAN are also secure because they also enable ACL.


แผนผังภายในมหาวิทยาลัย
(Campus Map)
ประตูทางเข้า-ออก
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ร้านสะดวกซื้อ 7-eleven
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อาคารบริการ
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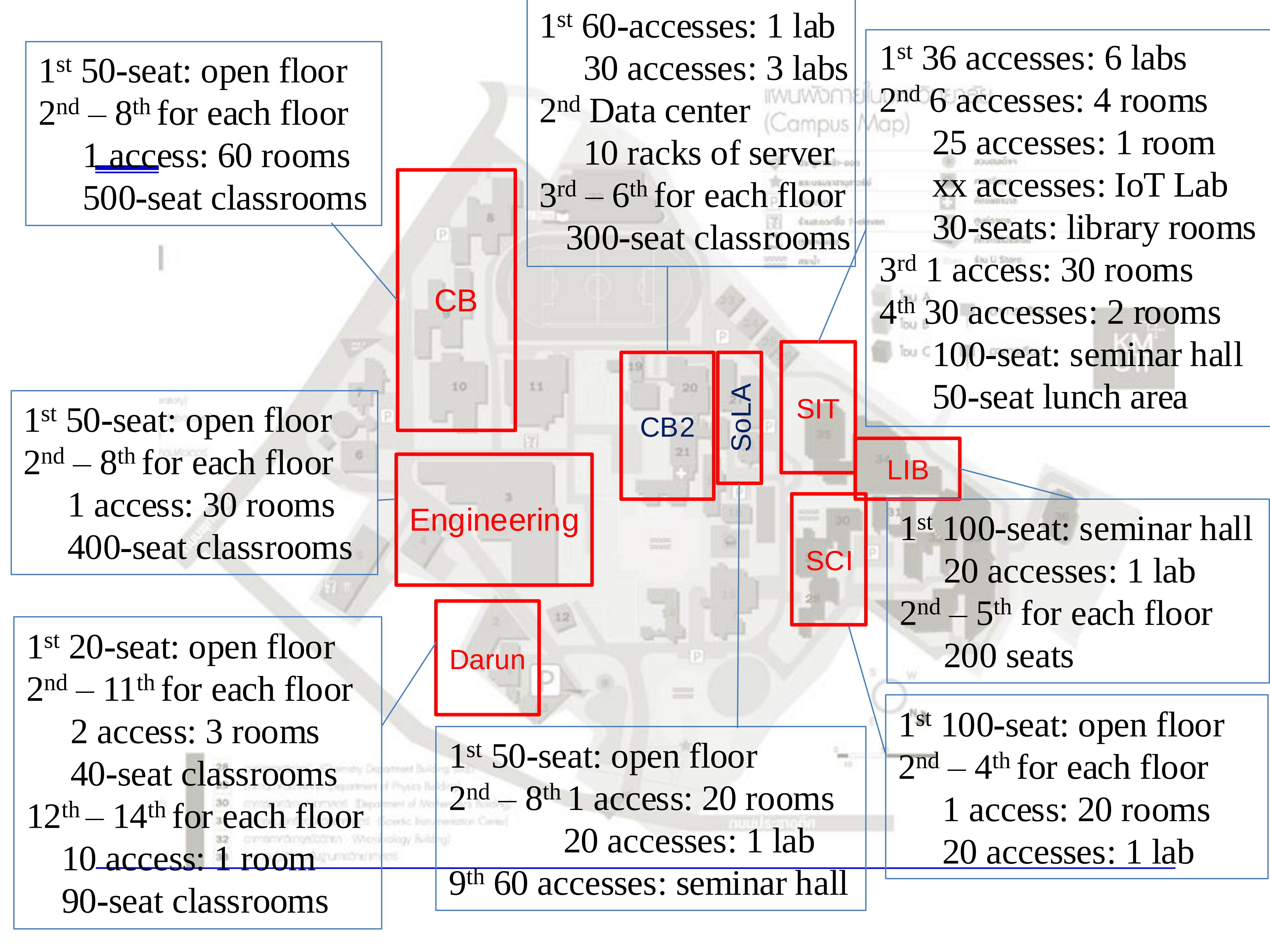
33 อาคารปฏิบัติการพื้นฐานทางวิทยาศาสตร์
KMUTT



1st 50-seat: open floor
2nd – 8th for each floor
1 access: 60 rooms
500-seat classrooms
1st 60-accesses: 1 lab
30 accesses: 3 labs
2nd Data center
10 racks of server
3rd – 6th for each floor
300-seat classrooms
1st 36 accesses: 6 labs
2nd 6 accesses: 4 rooms
25 accesses: 1 room
xx accesses: IoT Lab
30-seats: library rooms
3rd 1 access: 30 rooms
4th 30 accesses: 2 rooms
100-seat: seminar hall
50-seat lunch area
(Campus Map)
CB
CB2
SoLA
SIT
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Engineering
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200 seats
1st 20-seat: open floor
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2 access: 3 rooms
40-seat classrooms
12th – 14th for each floor
10 access: 1 room
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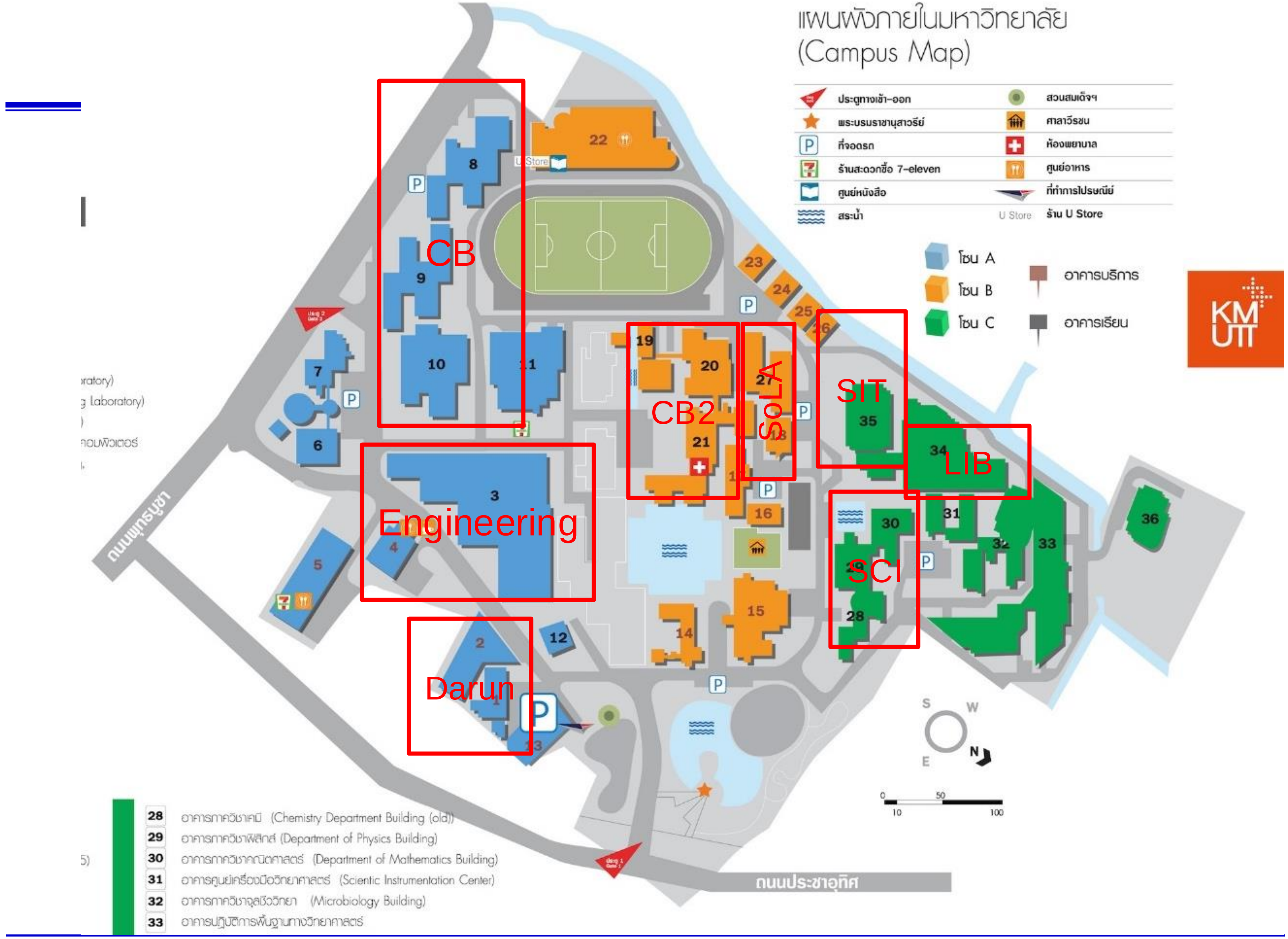

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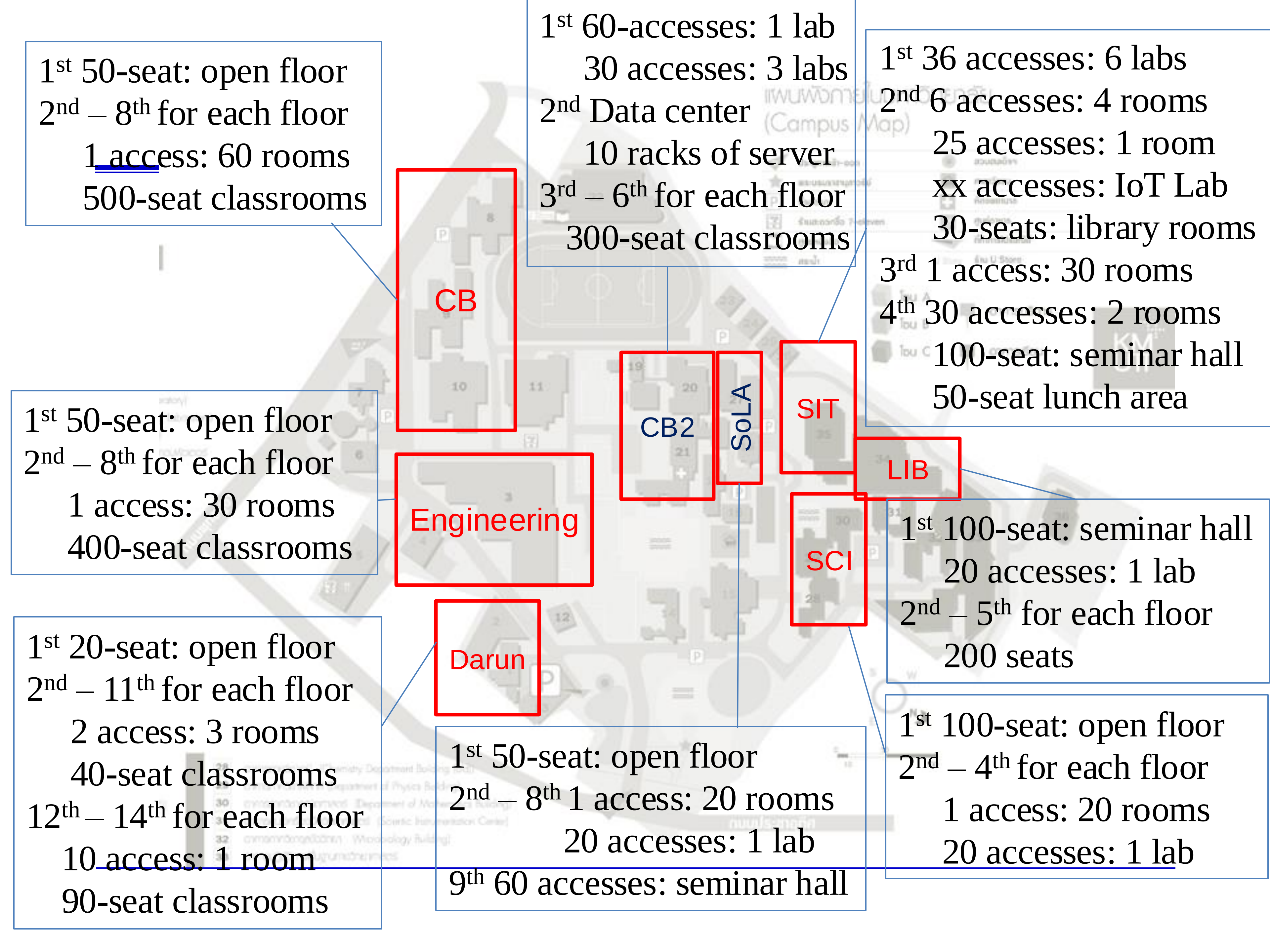

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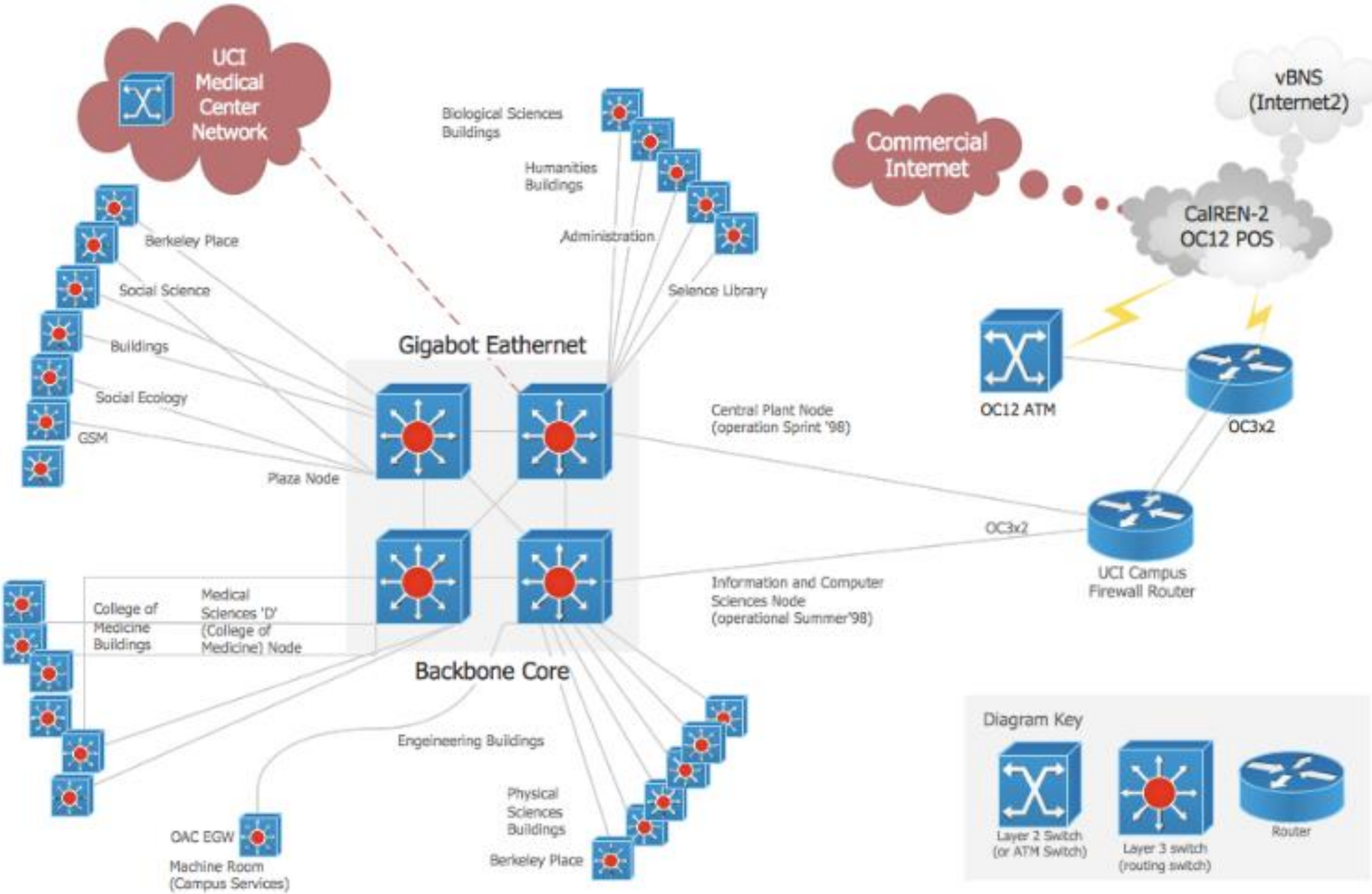
UCI Medical Center Network
Commercial Internet
vBNS (Internet2)
CalREN-2 OC12 POS
Biological Sciences Buildings
Humanities Buildings
Administration
Selence Library
Berkeley Place
Social Science
Buildings
Social Ecology
GSM
Plaza Node
Gigabot Eathernet
OC12 ATM
OC3x2
Central Plant Node (operation Sprint '98)
OC3x2
UCI Campus Firewall Router
College of Medicine Buildings
Medical Sciences 'D' (College of Medicine) Node
Information and Computer Sciences Node (operational Summer'98)
Backbone Core
Engeineering Buildings
Physical Sciences Buildings
Berkeley Place
OAC EGW
Machine Room (Campus Services)
Diagram Key
Layer 2 Switch (or ATM Switch)
Layer 3 switch (routing switch)
Router